

Homework 1

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Problem 1

Problem Statement

Let a and b be integers, Prove that if $2 + a$ and $35 - b$ are divisible by 11, then $a + b$ is divisible by 11.

Solution

Proof. Since $2 + a$ and $35 - b$ are divisible by 11, it can be represented with the following equations:

$$\begin{aligned}2 + a &= 11 \cdot p, & p &\in \mathbb{Z} \\35 - b &= 11 \cdot q, & q &\in \mathbb{Z}\end{aligned}$$

where p and q are integers.

Each equations can be rearranged as follows.

$$\begin{aligned}a &= 11 \cdot p - 2 \\b &= 35 - 11 \cdot q\end{aligned}$$

The sum $a + b$ is then as follows.

$$\begin{aligned}a + b &= 11p - 2 + 35 - 11q \\&= 11 \cdot (p - q) + 33 \\&= 11 \cdot (p - q) + 11 \cdot 3 \\&= 11 \cdot (p - q + 3) \\&= 11 \cdot k & k = p - q + 3 \in \mathbb{Z}\end{aligned}$$

k is guaranteed to be an integer since it is a sum of integers. Since $a + b$ is a product of 11 and some integer k , $a + b$ is divisible by 11.

□

Problem 2

Textbook p15 (15.c, 15.e)

Problem Statement

Use Euclidean algorithm to find:

(a) $(1003, 456)$

(b) $(5858, 1436)$

Solution

(a)

$$\begin{aligned}(1003, 456) &= (456, 91) \\ &= (91, 1) \\ &= (1, 0) \\ &= 1\end{aligned}$$

$$\begin{aligned}1003 &= 456 \times 2 + 91 \\ 456 &= 91 \times 5 + 1 \\ 91 &= 1 \times 91 + 0\end{aligned}$$

Answer: $(1003, 456) = 1$

(b)

$$\begin{aligned}(5858, 1436) &= (1436, 114) \\ &= (114, 68) \\ &= (68, 46) \\ &= (46, 22) \\ &= (22, 2) \\ &= (2, 0) \\ &= 2\end{aligned}$$

$$\begin{aligned}5858 &= 1436 \times 4 + 114 \\ 1436 &= 114 \times 12 + 68 \\ 114 &= 68 \times 1 + 46 \\ 68 &= 46 \times 1 + 22 \\ 46 &= 22 \times 2 + 2 \\ 22 &= 2 \times 11 + 0\end{aligned}$$

Answer: $(5858, 1436) = 2$

Problem 3

Textbook p15 (15.j)

Problem Statement

Express $(1003, 456)$ as a linear combination of 1003 and 456.

Solution

From using the Euclidean algorithm in Problem 2 (a), following equations are found.

$$1003 = 456 \times 2 + 91 \quad (1)$$

$$456 = 91 \times 5 + 1 \quad (2)$$

These equations can be rearranged as follows.

$$91 = 1003 - 456 \times 2 \quad (3)$$

$$1 = 456 - 91 \times 5 \quad (4)$$

By substituting Equation (3) to Equation (4), 1 can be represented as a linear combination of 1003 and 456.

$$\begin{aligned} 1 &= 456 - (1003 - 456 \times 2) \times 5 \\ &= 456 \times 11 - 1003 \times 5 \end{aligned} \quad (5)$$

The GCD of 1003 and 456 is 1, which can be represented as a linear combination of 1003 and 456 as follows.

$$1 = 456 \times 11 - 1003 \times 5$$

Problem 4

Problem Statement

Show that for any positive integer n , the fraction $\frac{12n+1}{30n+2}$ cannot be reduced.

Solution

Proof. For a fraction to be non-reducible, the denominator and the numerator must be coprime. In other words, the GCD of the denominator and the numerator must be 1.

The problem may be reformulated as follows: show that for any positive integer n , $(12n+1), 30n+2$ is 1. Note that since n is a positive integer, $12n+1$ is always less than $30n+2$.

Applying the Euclidean algorithm results in the following:

$$\begin{aligned} (30n+2, 12n+1) &= (12n+1, 6n) \\ &= (6n, 1) \\ &= (1, 0) \\ &= 1 \end{aligned}$$

From the above result, it is shown that the GCD of $30n + 2$ and $12n + 1$ is 1, meaning that $30n + 2$ and $12n + 1$ are coprime. Therefore, the fraction $\frac{12n+1}{30n+2}$ cannot be reduced for any positive integer n

□

Problem 5

Problem Statement

Find $(2^{100} - 1, 2^{120} - 1)$.

Solution

Let $q \in \mathbb{Z}^+$ be the GCD of $2^{100} - 1$ and $2^{120} - 1$. This can be represented as the following.

$$\begin{aligned}(2^{100} - 1) \bmod q &\equiv 0 \bmod q \\ (2^{120} - 1) \bmod q &\equiv 0 \bmod q\end{aligned}$$

By modulus arithmetic, the relations can be rearranged as following.

$$\begin{aligned}2^{100} \bmod q &\equiv 1 \bmod q \\ 2^{120} \bmod q &\equiv 1 \bmod q\end{aligned}$$

Then, the two relations can be combined.

$$2^{100} \bmod q \equiv 2^{120} \bmod q$$

Divide both sides by $2^{100} \bmod q$ and rearrange.

$$\begin{aligned}2^{20} \bmod q &\equiv 1 \bmod q \\ (2^{20} - 1) \bmod q &\equiv 0 \bmod q\end{aligned}$$

q must divide $2^{20} - 1$. The largest number that can achieve this is $2^{20} - 1$. Therefore, the GCD of $(2^{100} - 1, 2^{120} - 1)$ is $2^{20} - 1$.

Problem 6

Textbook p16 (29)

Problem Statement

Prove that a positive integer is divisible by 9 if and only if the sum of its digits is divisible by 9.

Solution

First, we prove that if a positive integer is divisible by 9, its sum of digits is divisible by 9.

Proof. Let $d_0, d_1, d_2, \dots, d_{n-1}$ be the digits of a n digit positive number k where $k = d_0 + 10^1 \cdot d_1 + 10^2 \cdot d_2 + \dots + 10^{n-1} \cdot d_{n-1}$. We would like to prove that if 9 divides k , 9 divides sum of $d_0, d_1, d_2, \dots, d_{n-1}$.

k can be reformulated as follows:

$$\begin{aligned} k &= d_0 + 10^1 \cdot d_1 + 10^2 \cdot d_2 + \dots + 10^{n-1} \cdot d_{n-1} \\ &= \sum_{i=0}^{n-1} 10^i \cdot d_i \\ &= \sum_{i=0}^{n-1} d_i + (10^i - 1) \cdot d_i \\ &= \sum_{i=0}^{n-1} d_i + \sum_{i=0}^{n-1} (10^i - 1) \cdot d_i \end{aligned}$$

By factoring $10^i - 1$, k can be further reformulated as follows:

$$\begin{aligned} k &= \sum_{i=0}^{n-1} d_i + \sum_{i=0}^{n-1} (10^i - 1) \cdot d_i \\ &= \sum_{i=0}^{n-1} d_i + \sum_{i=0}^{n-1} (10 - 1)(10^{i-1} + 10^{i-2} + \dots + 10^1 + 1) \cdot d_i \\ &= \sum_{i=0}^{n-1} d_i + 9 \sum_{i=0}^{n-1} (10^{i-1} + 10^{i-2} + \dots + 10^1 + 1) \cdot d_i \end{aligned}$$

Now k is in the form $k = ab + r$, where $r = \sum_{i=0}^{n-1} d_i$, $a = 9$, and $b = \sum_{i=0}^{n-1} (10^{i-1} + 10^{i-2} + \dots + 10^1 + 1) \cdot d_i$. Since 9 divides k , 9 also must divide r , which is the sum of digits. Therefore, if a positive integer is divisible by 9, sum of its digits are also divisible by 9. $\square$

Next, we prove that if the sum of a positive integer's digits is divisible by 9, the positive integer is also divisible by 9.

Proof. From previous formulation of k , it is established that $k = ab + r$ where $a = 9$. Since r is the sum of digits and can be represented in the form $r = 9p$ for some integer p , k can be reformulated as follows:

$$\begin{aligned} k &= ab + r \\ &= 9b + 9p \\ &= 9(b + p) \end{aligned}$$

This shows that k is a multiple of 9, meaning that it is divisible by 9. $\square$

Problem 7

Textbook p16 (31.b)

Problem Statement

If t is an integer such that $a_i \mid t$ for $i = 1, 2, \dots, k$, prove that $[a_1, a_2, \dots, a_k] \mid t$.

Solution

Proof. Let $m = [a_1, a_2, \dots, a_k]$. By division algorithm, $t = mq + r$. For t to be divisible by m , r must be 0.

For every i , let $t = a_i u_i$ and $m = a_i v_i$ since t and m are divisible by a_i . By rearranging the equation found by division algorithm, $r = t - mq$. Substituting t and m to this equation results in the following: $r = a_i(u_i - v_i q)$. Therefore, r is divisible by a_i for all i $\square$

Problem 8

Textbook p23 (21)

Problem Statement

If $c^2 = ab$ and $(a, b) = 1$, where a, b, c are positive integers, prove that a and b are perfect squares.

Solution

Proof. Assume contradiction: if $c^2 = ab$ and $(a, b) = 1$, then a and b are not perfect squares.

From the second equation, it can be deduced that a and b are coprime, meaning that a does not equal to b .

Square rooting the first equation yields the following equation: $c = \sqrt{ab} = \sqrt{a}\sqrt{b}$. Since a does not equal to b , a and b multiplied cannot result in a perfect

square. However, since a and b are positive integers, square root of them is irrational. This is a contradiction, as c must also be a positive integer.

Therefore, by contradiction, if $c^2 = ab$ and $(a, b) = 1$, a and b are perfect squares. $\square$

Problem 9

Textbook p24 (25)

Problem Statement

Let p be prime and $1 \leq k < p$. Prove that p divides the binomial coefficient $\binom{p}{k}$.

Solution

Proof. Recall definition of binomial coefficient.

$$\binom{p}{k} = \frac{p!}{k!(p-k)!}$$

Rearrange the equation.

$$p! = \binom{p}{k} k!(p-k)!$$

The left-hand side is divisible by p . By Euclid's Lemma, one of two terms $\binom{p}{k}$ and $k!(p-k)!$ must be divisible by p . However, the second term is not divisible by p , since $k < p$ and $p-k < p$. Therefore, $\binom{p}{k}$ must be the term divisible by p . $\square$

Problem 10

Textbook p24 (32)

Problem Statement

Prove that there are infinitely many primes.

Solution

Proof. Assume that there are finitely many primes. Let these primes be $p_1, p_2, \dots, p_k$.

Consider $n = p_1 p_2 \dots p_k + 1$. By division algorithm, $n = mq + r$ for some integer m where $q = p_1 p_2 \dots p_k$ and $r = 1$. This means that n is not divisible by p_i for all i since $r \neq 0$. Since p_i are primes for all i , it does not have any factors besides 1 and itself. By extension, all integers less than n cannot divide n . Therefore, n is a prime. This is a contradiction.

The above logic applies to any sets of prime by induction. Therefore, there are infinitely many primes. $\square$

Problem 11

Problem Statement

Solution

Problem 12

Problem Statement

Solution